=== FORMULAS DIRECTLY EXTRACTED FROM USER PDF ===

Planned Value (PV)

PV = (Planned % Complete) × BAC

Earned Value (EV)

EV = (% Completed) × BAC

Actual Cost (AC)

(Directly taken from project records)

Schedule Variance (SV)

SV = EV – PV

Cost Variance (CV)

CV = EV – AC

Schedule Performance Index (SPI)

SPI = EV / PV

Cost Performance Index (CPI)

CPI = EV / AC

Forecasting (Extracted from PDF)

ETC (Performance-based)

ETC = (BAC – EV) / CPI

EAC (Several formulas)

EAC = AC + (BAC – EV)

EAC = AC + (BAC – EV) / CPI

EAC = AC + (BAC – EV) / (CPI × SPI)

VAC

$VAC = BAC - EAC$

TCPI (Under Budget)

$TCPI = (BAC - EV) / (BAC - AC)$

TCPI (Over Budget, using EAC)

$TCPI = (BAC - EV) / (EAC - AC)$

=== ADDED CONTENT (NOT IN PDF): NPV + EVM INTEGRATION ===

Net Present Value (NPV)

$NPV = \Sigma [Ct / (1 + r)^t] - C0$

EVM Adjustments for NPV Integration

Cost Adjustment using CPI:

$Adjusted\_Cost\_t = Ct / CPI$

Schedule Adjustment using SPI:

$Adjusted\_Time = t / SPI$

Integrated NPV-EVM Formula:

$NPV = \Sigma [(Ct / CPI) / (1 + r)^{(t / SPI)}] - C0$

Purpose:

This integration helps forecast true financial value while considering project cost efficiency (CPI) and schedule efficiency (SPI).